

PROGRAMMING FOR PROBLEM SOLVING LAB

I - II Semester: Common to all branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES:

- 1) To be familiarize with flow algorithm to solve simple problems
- 2) To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
- 3) To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.

LIST OF EXPERIMENTS

Week-1 INTRODUCTION TO FLOWGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle
9. Draw a flowchart to check whether a given number is palindrome or not.

Week-2 BASIC DATA TYPES

- 1) Write a C program to find division of 1 integer 1 float numbers.
- 2) Write a C program to find division of 2 integer numbers.
- 3) Write a C program to find average of (1int,1float) numbers.
- 4) Write a C Program to Swap Numbers without Using Temporary Variable
- 5) Write a C Program to Swap two Numbers Using Temporary Variable
- 6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.
 - a) $(x+y+z) / (x-y-z)$
 - b) $(x+y+z) / 3$
 - c) $(x+y) * (x-y) * (y-z)$

Week-3 OPERATORS

- 1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ($c = (f - 32) / 1.8$)
- 2) Write a C program to find area and perimeter of a circle. ($\text{area} = \pi r^2$ $\text{perimeter} = 2\pi r$)
- 3) Write a C program to calculate area and perimeter of a right angled triangle.
 - a. ($\text{Area} = 1/2 * b * h$ $\text{perimeter} = w + h + \sqrt{w^2 + h^2}$)
- 4) Find the sum of natural numbers 1 to n. (read n as input) (Use formula $\text{sum} = n(n+1)/2$)
- 5) Write a C program to calculate Simple interest ($\text{SI} = \text{PTR}/100$)
- 6) Write a C program to calculate area and perimeter of a rectangle. $\text{Area} = l * b$ $\text{Perimeter} = 2 * (l + b)$
- 7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input. ($a + b + c = 180$)
- 8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ($\text{Amt} = \text{num of units} * \text{cost}$)
- 9) Write a C Program to calculate area and perimeter of a triangle.
 - a. $\text{Perimeter} = (a + b + c)$
 - b. $s = (a + b + c) / 2$
 - c. $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$
- 10) Write a C program to read five Subject marks and find the average.
- 11) Write a C program to Calculate Compound interest ($\text{CI} = p(1 + r/100)^n$)

Week-4	CONDITIONAL STATEMENTS
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1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator from the user (+, -, *, /, %) use switch)
3. Write a program to compute grade of students using if else ladder. The grades are assigned as followed:

$\text{marks} < 50$ F
 $50 \leq \text{marks} < 60$ C
 $60 \leq \text{marks} < 70$ B
 $70 \leq \text{marks}$ B+
 $80 \leq \text{marks} < 90$ A
 $90 \leq \text{marks} \leq 100$ A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

Week-5	LOOPING STATEMENTS
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- 1) Write a C program to find Sum of individual digits of given integer
- 2) Write a C program to generate first n terms of Fibonacci series
- 3) Write a C Program to find the Sum of Series $\text{SUM} = 1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$
- 4) Write a C program to print the Fibonacci sequence up to given value of n.
- 5) Write a C program to print the multiplication table to the given value of n.
- 6) Write a C program to check whether given number is palindrome or not
- 7) Write a C program to check whether given number is perfect or not

Week-6	NESTED LOOPING STATEMENTS
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- 1) Write a C program to generate prime numbers between 1 and n
- 2) Write a C program to generate Pascal's triangle.
- 3) Write a C program to generate the following pyramid of numbers.

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      1
    1 3 1
  1 3 5 3 1

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- 4) Write a C program to generate the pattern

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Week-7**ARRAYS**

- 1) Write a C Program to implement following searching methods
 - i. Binary Search
 - ii. Linear Search
- 2) Write a C program to find largest and smallest number in a list of integers
- 3) Write a C program
 - i. To add two matrices
 - ii. To multiply two matrices
- 4) Write a C program to find Transpose of a given matrix

Week-8**FUNCTIONS**

- 1) Write a C program to find the factorial of a given integer using non recursive functions
- 2) Write a C program to find GCD of given integers using non recursive functions
- 3) Write a C Program to find the power of a given number using non recursive functions
- 4) Write a C program to find sum of natural numbers using non recursive function.
- 5) Write a C program to reverse a given integer number using non recursive functions
- 6) Write a C Program to find binary equivalent of a given decimal number using recursive functions.
- 7) Write a C Program to print Fibonacci sequence using recursive functions.
- 8) Write a C Program to find LCM of 3 given numbers using recursive functions
- 9) Write a C program to find the factorial of a given integer using recursive functions
- 10) Write a C program to print fibonacci series till n terms using recursion.

Week-9**STRINGS**

- 1) Write a C program using to Insert a sub string into a given main string from a given position
- 2) Write a C program using to Delete n characters from a given position in a string
- 3) Write a C program to determine if given string is palindrome or not
- 4) Write C Programs to demonstrate the following string handling functions.
 - a. strcat()
 - b. strcmp()
 - c. strrev()
 - d. strcpy()
 - e. strlen()
 - f. strstr()
 - g. strncpy()
 - h. strncat()
 - i. strncmp()

Week-10	POINTERS
1) Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers. 2) Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference. 3) Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array. 4) Write a C Program to print 2-D array using pointers.	
Week-11	STRUCTURES
1) Write a C Program using functions to - Reading a complex number - Writing a complex number - Add two complex numbers - Multiply two complex numbers Note: represent complex number using structure 2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.	
Week-12	FILES
1) Write a C program to read and print the content of a file. 2) Write a C program copy the content of one file to another file 3) Write a C program to merge two file into third file. 4) Write a C Program to find the number of lines in a text file	
Text Books:	
1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27. 2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13 th Edition, 2012.	
Reference Books:	
1) Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018 2) King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2 nd Edition, 2015. 3) Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3 rd Edition, 2004. 4) Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1 st Edition, 1994.	
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO: At the end of the course, student will be able to - Solve simple mathematical problems using Flowgorithm. - Correct syntax errors as reported by the compilers and logical errors encountered at run time. - Develop programs by using decision making and looping constructs. - Implement real time applications using the concept of array, pointers, functions and structures. - Solve real world problems using matrices, searching and sorting	